PROJECT ORBIT

10-Minute Presentation Guide *— slides, speaking script, architecture and demo plan*

Project: ORBIT – E-Commerce Microservices Platform

Presentation objective: Explain what the project does, how the backend and frontend are divided, and demonstrate a few working screens.

1. Project Description

Two-line description:

“Orbit is an e-commerce application providing an end-to-end shopping experience, from product browsing and cart management to order, payment and notification handling.”

“The application follows a microservices architecture where business capabilities such as authentication, products, inventory, orders, payments and notifications are separated into focused services.”

Technology: Java • Spring Boot • REST APIs • JPA/Hibernate • PostgreSQL • React • JavaScript • JWT • API Gateway • Docker • Git/GitLab • Maven • Postman

2. Architecture

React Frontend → API Gateway :9191 → Auth / Product / Order / Inventory / Payment / Notification Services → Dedicated PostgreSQL databases

Speaking line: “The frontend communicates through the API Gateway. Behind it, each service focuses on a specific business capability and maintains database separation.”

3. Backend Modules

Authentication Service — Registration, login, password encryption, JWT generation and role-based access.

Product Service — Product CRUD, category, price, image and product details.

Inventory Service — Stock management, availability checks and order-driven stock updates.

Order Service — Order creation, order items, totals, confirmation, processing and cancellation.

Payment Service — Payment processing, success/failure states and refund handling.

Notification Service — Order/payment notifications, status and read/unread behavior.

4. Frontend Modules

Authentication — Login, registration and protected routes.

Product Catalog — Product listing, details, image, price, stock status and Add to Cart.

Shopping Cart — Quantity controls, remove item and total calculation.

Orders & Payment — Checkout, order status, payment result and refund flow.

Notifications — Unread count, read/unread state and mark-as-read behavior.

Admin Dashboard — Product, inventory and order management.

5. End-to-End User Flow

Login → Browse Products → Add to Cart → Checkout → Create Order → Inventory Check → Payment → Order Confirmed → Notification

Keep this section conversational; explain internals only if the panel asks.

6. Screenshot / Demo Plan

01. Login — Show the authentication screen.

02. Product Catalog — Show product image, name, price, category/stock and Add to Cart.

03. Cart / Checkout — Show quantities, subtotal/total and checkout.

04. Admin Dashboard — Show Products, Inventory and Orders.

05. Notifications — Show unread count; click a notification and show the count decreasing.

7. 10-Minute Timing

1.5 min Project description **1 min** Architecture **2 min** Backend **1.5 min** Frontend **1 min** Flow **2 min** Demo **1 min** Closing

8. Exact Opening Script

“Good morning. Today I’ll give a brief overview of my project, Orbit, an e-commerce application built using a microservices architecture. The goal is to provide an end-to-end shopping experience, from authentication and product browsing to cart, order, payment and notifications. I’ll first explain the architecture, then the backend and frontend modules, and finally show a few screens from the application.”

9. Exact Closing Script

“To summarize, Orbit demonstrates an end-to-end e-commerce workflow using independently focused microservices. The application combines a React user experience with secured Spring Boot REST services, PostgreSQL databases and integrations across order, inventory, payment and notification capabilities. Thank you.”

Final tip: Replace the five screenshot placeholders in the PPT with your real Orbit screens before presenting.